

3.1 Newton's Laws of Motion

Question Paper

Course	CIEA Level Physics
Section	3. Dynamics
Topic	3.1 Newton's Laws of Motion
Difficulty	Hard

Time allowed: 50

Score: /40

Percentage: /100

Question 1a

A person stands on weighing scales in a lift, as shown in Fig. 1.1. The scales are calibrated to give readings in newtons. It reads 700 N when the lift is stationary.

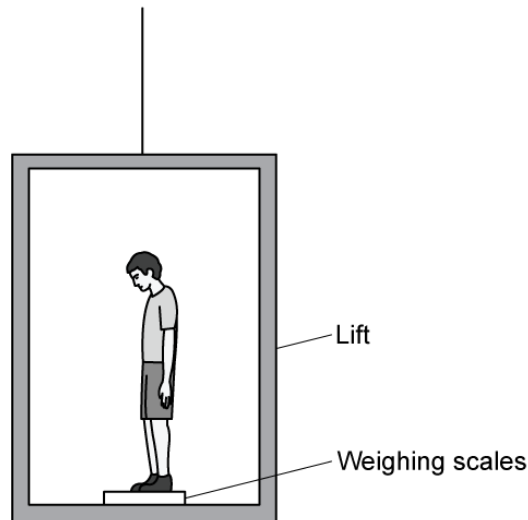


Fig. 1.1

When the lift starts to move, it accelerates for a short time. It continues to move upwards at a constant speed and eventually starts to decelerate until it comes to rest.

- (i)
Draw and label free-body force diagrams for the person during the periods of acceleration and deceleration.

[2]

- (ii)
By applying one of Newton's laws of motion, determine an expression for the apparent weight of the person during the deceleration.

[1]

- (ii)
By applying one of Newton's laws of motion, determine an expression for the apparent weight of the person during the acceleration.

[1]

[4 marks]

Question 1b

The magnitude of the lift's acceleration is 0.7 m s^{-2} .

On Fig. 1.2, sketch the expected readings on the weighing scales during acceleration and deceleration of the lift.

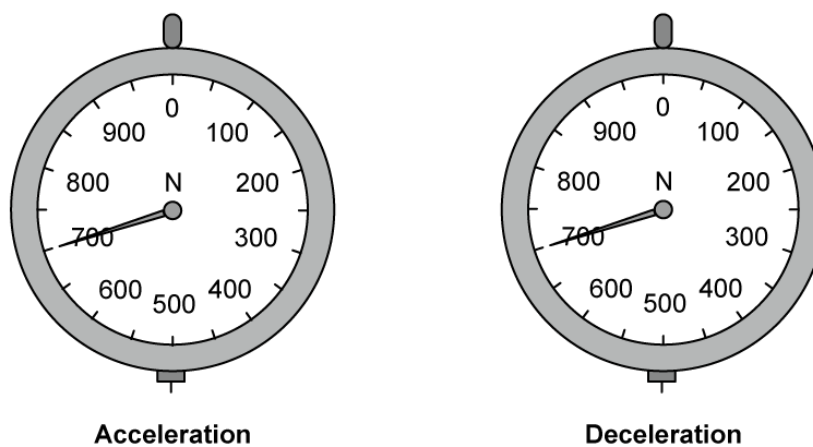


Fig. 1.2

[3 marks]

Question 1c

Fig. 1.3 shows the variation of acceleration with time for the duration of the journey in the lift.

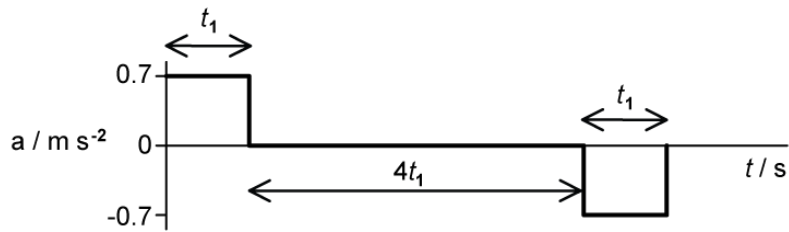


Fig. 1.3

The total height ascended by the lift is 24 m.

Determine the value of t_1 .

[2 marks]

Question 1d

On Fig. 1.4, sketch the variation of time with the person's apparent weight over the course of the motion.

Include values on the axes.

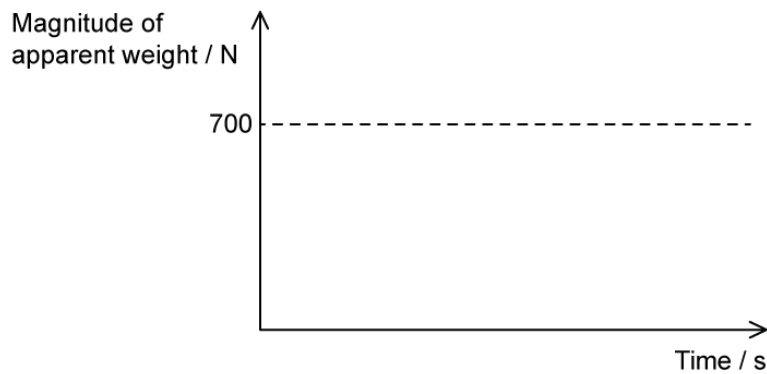


Fig. 1.4

[3 marks]

Question 2a

Three boxes A, B, and C are connected using an inextensible rope of negligible mass, as shown in Fig. 1.1.

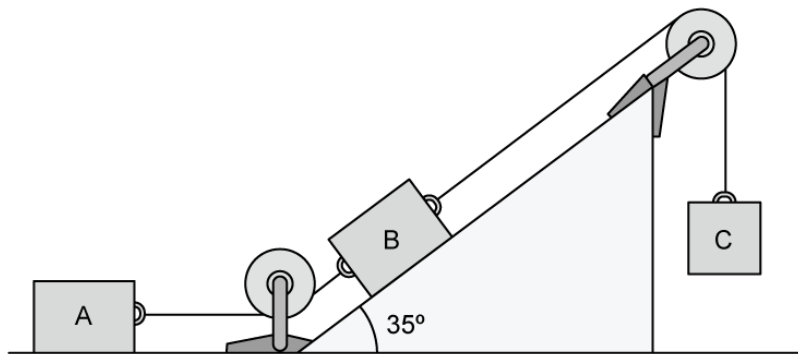


Fig. 1.1

Boxes A and B each have a mass of 2.4 kg, and the magnitude of the frictional force F between each box and the surface is equal to

$$F_x = \frac{1}{3} R_x$$

where R is the normal reaction force and x refers to the box in question A, B or C.

Box C descends with constant velocity.

Calculate the tension in the rope connecting boxes A and B.

[3 marks]

Question 2b

Determine the tension in the rope connecting boxes B and C.

[4 marks]**Question 2c**

Hence, determine the mass of box C.

[1 mark]**Question 2d**

The rope connecting A and B breaks abruptly.

Calculate the acceleration of block C.

[4 marks]

Question 3a

A rock climber of mass 80 kg slips after attaching a bolt firmly onto a piece of rock that they are climbing as shown in Fig. 1.1.

At the lowest point of the climber's fall, the length of the rope between the harness and the bolt is 24 m. The climber comes to rest and all the energy from the fall has been absorbed by the rope.

The rope has an unstretched length of 19 m and a stiffness of 560 N m^{-1} .

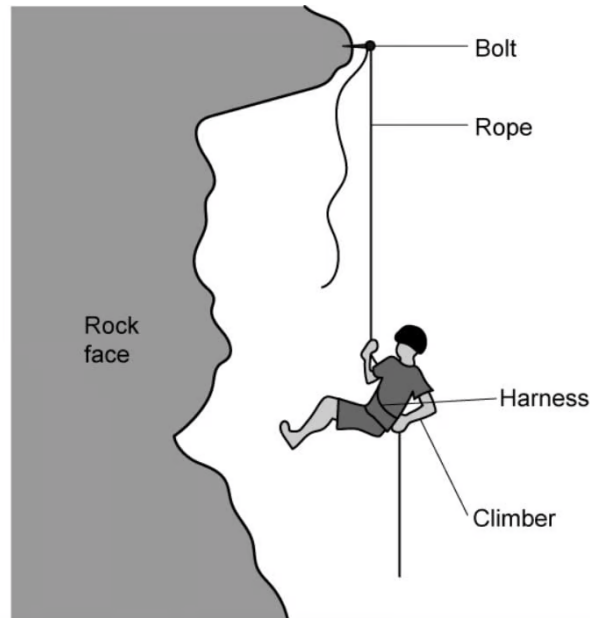


Fig. 1.1

Assume that the extension of the rope is proportional to its tension and the mass of the rope and the effects of air resistance are negligible.

Calculate

(i)
the resultant force acting on the climber when he reaches the lowest point in the fall.

[3]

(ii)
the extension of the rope when the climber's acceleration is zero.

[2]

[5 marks]

Question 3b

(i)

Use the principle of conservation of energy to calculate the maximum speed of the climber before the rope begins to stretch.

[2]

(ii)

Suggest whether using a longer rope would reduce the force experienced by the climber when the rope begins to stretch.

[2]

[4 marks]

Question 3c

Hemp is one of the strongest fibres with a very high tensile strength and is extremely rigid. Hemp ropes were the first to be used for ship rigging which adjusted the position of the sails and supported the masts.

However, rock climbers use nylon ropes to break their fall as they can stretch considerably under tension.

A climbing equipment company tests their ropes by attaching an 80 kg mass to an end of the rope and releasing it from a ledge.

Fig. 1.2 shows the results of a test carried out to compare the time it takes for a nylon rope and a hemp rope to return to their unstretched length after beginning to stretch.

Fig. 1.2

Rope	Time to return to unstretched length / s
Nylon	0.664
Hemp	0.035

In the test, both ropes used had identical unstretched lengths of 19.0 m and diameters of 10 mm.

Show that the average resultant force exerted by the hemp rope is about 20 times larger than the average resultant force exerted by the nylon rope.

[3 marks]

Question 3d

Discuss in terms of force and momentum, why nylon ropes are favoured by rock climbers compared to hemp ropes.

[4 marks]

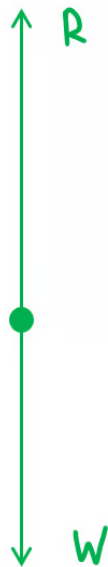
Model Answers: Hard

1a

(a)

(i) Sketch free-body force diagrams for the periods of acceleration and deceleration:

During acceleration:



- Contact force $>$ weight i.e. the contact force is larger than the weight; [1 mark]

During deceleration:



- Contact force < weight i.e. the contact force is smaller than the weight; [1 mark]

(ii) The apparent weight of the person during acceleration is:

- Newton's Second Law: $F = ma$
- $F = R - W$
- $R = mg + ma$ [1 mark]

(iii) The apparent weight of the person during deceleration is:

- $F = W - R$
- $R = mg - ma$ [1 mark]

[Total: 4 marks]

Imagine the two forces acting on the person: one upward force due to the normal reaction force from standing on the scale, and one downward force due to their weight.

The weight of the person **is always constant**, but there will be a **resultant force** acting upwards or downwards, depending on the direction of the acceleration. The **apparent weight** is therefore referring to the size of the **contact force** experienced by the person as this is the only force that changes.

When drawing free body diagrams, make sure that the size of the weight arrow stays the same size in both diagrams!

1b

(b) Sketch the expected readings on the scales:

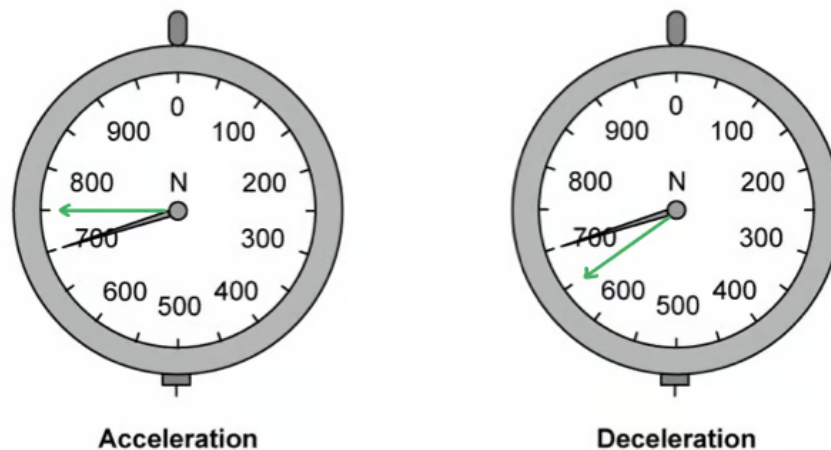
List the known quantities:

- Acceleration, $a = 0.7 \text{ m s}^{-2}$
- Weight of the person, $W = 700 \text{ N}$

Calculate the resultant force:

- Newton's Second Law: $F = ma$
- $F = \left(\frac{700}{9.81} \right) \times 0.7$
- $F = 49.9 = 50 \text{ N}$
- Therefore, reading during acceleration = $700 + 50 = 750 \text{ N}$ [1 mark]
- Therefore, reading during deceleration = $700 - 50 = 650 \text{ N}$ [1 mark]

The readings on the scale would be as follows:



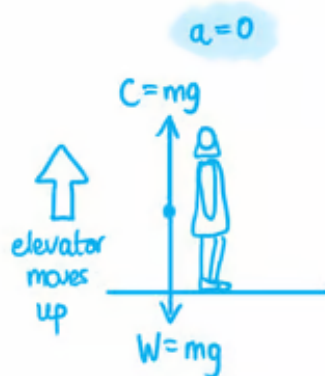
- Readings from calculation correctly marked with a straight line from the centre of the dial; [1 mark]

[Total: 3 marks]

When the lift accelerates, the **upward contact force** increases. This is interpreted by the weighing scales as an increased reading.

When the lift decelerates, it's the opposite. Try to imagine a lift moving upward but coming to a stop (i.e., **decelerating**) as it approaches the floor above. You will feel momentarily a bit 'floaty' like you are about to lift off of the floor. This experience is because the upward **contact force** from the floor on your feet **decreases** as the lift slows down.

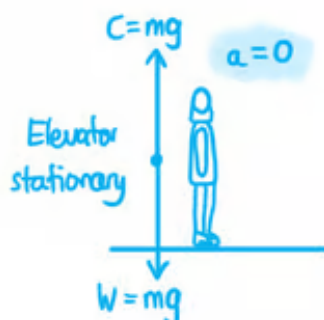
In this case, there must be a resultant force **downwards** (because the lift is slowing down) but the weight of the person stays constant. Hence, the contact force decreases (and the reading on the spring scale would be **less** than 700 N).



While the elevator moves with a constant velocity, the contact force $C = mg$. This is due to Newton's 3rd Law (the person exerts a force mg on the floor!)



While the elevator slows down, the contact force $C < mg$ because there is a resultant force downwards. Note, the force exerted on the floor is also less than mg (again, Newton's 3rd Law!)

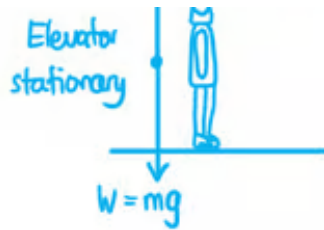


While the elevator is at rest, the contact force must equal mg again, because there is no resultant force.

1c

(c) Determine the value of t_1 :

- Distance travelled (height) = average velocity of each segment $\times$ time
- Distance travelled: $(0.7t_1) \times t_1 + (0.7t_1) \times 4t_1 + (0.7t_1) \times t_1 = 24 \text{ m}$ [1 mark]
- Distance travelled: $0.7t_1^2 + 2.8t_1^2 + 0.7t_1^2 = 24 \text{ m}$



contact force must equal weight,
because there is no resultant force.

1c

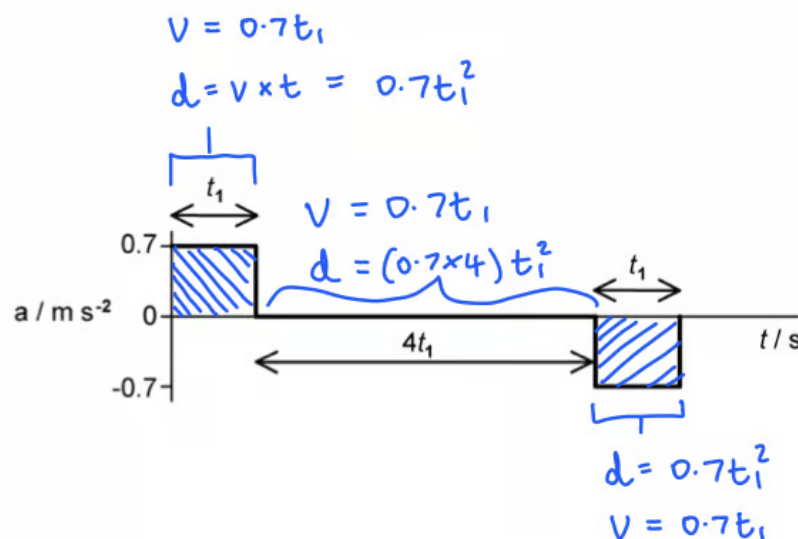
(c) Determine the value of t_1 :

- Distance travelled (height) = average velocity of each segment $\times$ time
- Distance travelled: $(0.7t_1) \times t_1 + (0.7t_1) \times 4t_1 + (0.7t_1) \times t_1 = 24 \text{ m}$ [1 mark]
- Distance travelled: $0.7t_1^2 + 2.8t_1^2 + 0.7t_1^2 = 24 \text{ m}$
- Simplify and rearrange: $4.2t_1^2 = 24 \text{ m} \Rightarrow t_1 = \sqrt{\frac{24}{4.2}}$
- $t_1 = 2.39 = 2.4 \text{ s}$ [1 mark]

[Total: 2 marks]

Consider the equation for acceleration:

$$a = \frac{\Delta v}{t} \text{ so } \Delta v = at$$



During initial constant acceleration:

- The lift accelerates from rest at 0.7 m s^{-2} for t_1 seconds
- Hence, average velocity = $0.7t_1 \text{ m s}^{-1}$

During the period of constant velocity:

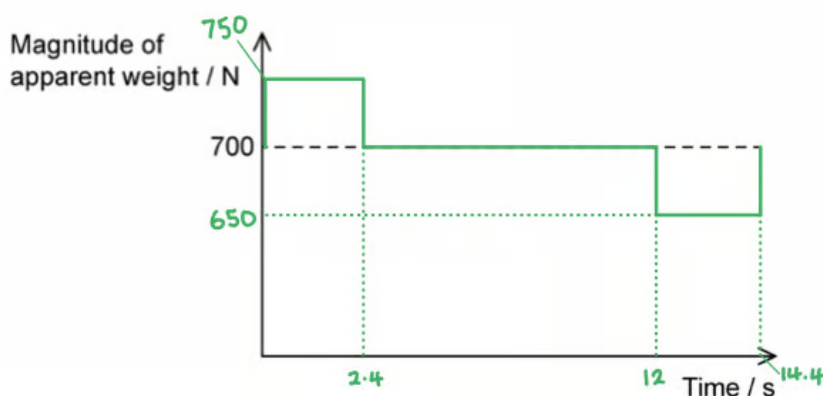
- The lift continues at $0.7t_1 \text{ m s}^{-1}$ for $4t_1$ seconds
- Hence, (average) velocity = $0.7t_1 \text{ m s}^{-1}$

During the final constant deceleration:

- The lift decelerates at -0.7 m s^{-2} for t_1 seconds back to zero velocity
- Hence, average velocity = $-0.7t_1 \text{ m s}^{-1}$

1d

(d) The variation of time with the apparent weight for the journey:



- Same shape as the acceleration-time graph in (c); [1 mark]
- Values for apparent weight correctly marked; [1 mark]
- Values for time correctly marked (values or multiples of t_1); [1 mark]

[Total: 3 marks]

The key to figuring out the shape of this graph is to realise that $F \propto a$, so the shape will be exactly the same as the acceleration graph in the previous question.

You have already calculated the magnitude of apparent weight in part (b), so don't calculate them again!

2a

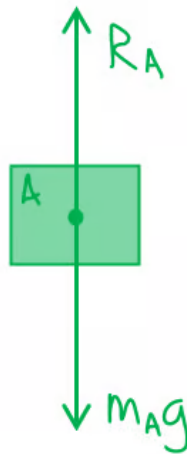
(a) Calculate the tension in the rope connecting blocks A and B:

List the known quantities:

- Frictional force on A, $F_A = \frac{1}{3}R_A$
- Mass of A, $m_A = 2.4 \text{ kg}$
- Constant velocity, so acceleration $a = 0$

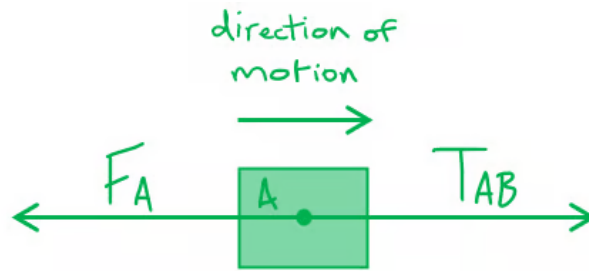
Consider the vertical forces acting on box A:

- Resultant force: $R_A - m_A g = 0$
- Therefore, $R_A = m_A g = 2.4 \times 9.81 = 23.5 \text{ N}$ [1 mark]
- Hence frictional force: $F_A = \frac{1}{3} \times 23.5 = 7.8 \text{ N}$ [1 mark]



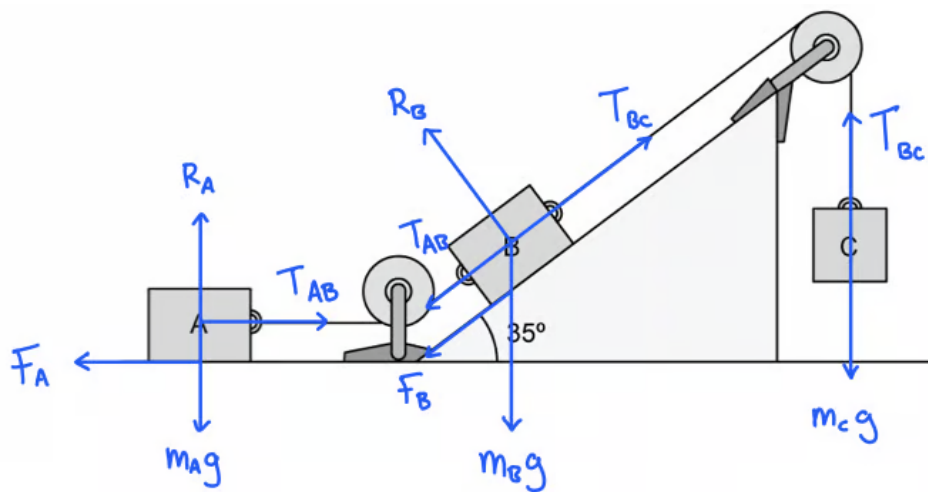
Consider the horizontal forces acting on box A:

- Resultant force: $T_{AB} - F_A = 0$
- Therefore, $T_{AB} = F_A = 7.8 \text{ N}$ [1 mark]



[Total: 3 marks]

It's always best to spend some time drawing the free-body diagram first before jumping into any calculations



2b

(b) Determine the tension in the rope connecting boxes B and C:

List the known quantities:

- Frictional force on B, $F_B = \frac{1}{3} R_B$
- Mass of B, $m_B = 2.4 \text{ kg}$
- Constant velocity, so acceleration $a = 0$
- Tension in rope AB, $T_{AB} = 7.8 \text{ N}$ (from **(a)**)
- Angle of inclination of ramp, $\theta = 35^\circ$

Consider the forces acting perpendicular to the slope on box B:

- Resultant force: $R_B - m_B g \cos 35^\circ = 0$

- Therefore, $R_B = m_B g \cos 35^\circ = 2.4 \times 9.81 \times \cos 35^\circ = 19.3 \text{ N}$ [1 mark]
- Hence frictional force: $F_B = \frac{1}{3} \times 19.3 = 6.4 \text{ N}$ [1 mark]

Consider the forces acting parallel to the slope on box B:

- Resultant force: $T_{BC} - T_{AB} - m_B g \sin 35^\circ - F_B = 0$ [1 mark]

$$\therefore T_{BC} = T_{AB} + m_B g \sin 35^\circ + F_B$$

Consider the forces acting perpendicular to the slope on box B:

- Resultant force: $R_B - m_B g \cos 35^\circ = 0$
- Therefore, $R_B = m_B g \cos 35^\circ = 2.4 \times 9.81 \times \cos 35^\circ = 19.3 \text{ N}$ [1 mark]
- Hence frictional force: $F_B = \frac{1}{3} \times 19.3 = 6.4 \text{ N}$ [1 mark]

Consider the forces acting parallel to the slope on box B:

- Resultant force: $T_{BC} - T_{AB} - m_B g \sin 35^\circ - F_B = 0$ [1 mark]
- $T_{BC} = T_{AB} + m_B g \sin 35^\circ + F_B$
- $T_{BC} = 7.8 + (2.4 \times 9.81 \times \sin 35^\circ) + 6.4$
- $T_{BC} = 27.7 \text{ N}$ [1 mark]

[Total: 4 marks]

2c

(c) Determine the mass of box C:

- Box C descends with a constant velocity, so resultant force: $T_{BC} - m_C g = 0$
- Therefore, $T_{BC} = m_C g = 27.7 \text{ N}$
- Mass of C: $m_C = \frac{27.7}{9.81} = 2.8 \text{ kg}$ [1 mark]

[Total: 1 mark]

2d

(d) Calculate the acceleration of block C:

List the known quantities:

- Mass of B, $m_B = 2.4 \text{ kg}$
- Mass of C, $m_C = 2.8 \text{ kg}$ (from **(c)**)
- Weight of C, $m_C g = 27.7 \text{ N}$ (from **(c)**)
- Frictional force on B, $F_B = 6.4 \text{ N}$ (from **(b)**)
- New tension in BC = T

Determine the new resultant forces on B and C:

- Resultant force on B: $T - m_B g \sin 35^\circ - F_B = m_B a$ **[Eq. 1]** [1 mark]
- Resultant force on C: $m_C g - T = m_C a$ **[Eq. 2]** [1 mark]

Add [Eq. 1] and [Eq. 2] together and rearrange for a :

- $T - m_B g \sin 35^\circ - F_B + m_C g - T = m_B a + m_C a$
- $m_C g - m_B g \sin 35^\circ - F_B = (m_B + m_C) a$
- $a = \frac{m_C g - m_B g \sin 35^\circ - F_B}{m_B + m_C}$ [1 mark]

Substitute in the known quantities and calculate:

- $a = \frac{27.7 - (2.4 \times 9.81 \times \sin 35^\circ) - 6.4}{2.4 + 2.8}$
- $a = 1.499 = 1.5 \text{ m s}^{-2}$ [1 mark]

[Total: 4 marks]

There is a change in the tension in the rope between B and C because the forces acting on block B have changed.

3a

(a)

List the known quantities:

- Mass of the climber, $m = 80 \text{ kg}$
- Stretched length of rope = 24 m
- Unstretched length of rope, $L = 19 \text{ m}$
- Stiffness of rope, $k = 560 \text{ N m}^{-1}$

(i) Calculate the resultant force acting on the climber when he reaches the lowest point in the fall:

- The lowest point is when the rope reaches its maximum extension
- $\Delta x_{\text{max}} = \text{stretched length} - \text{unstretched length} = 24 - 19 = 5 \text{ m}$ [1 mark]

Determine the resultant force on the climber:

- Resultant = $F - W$ where $F = k \Delta x$ (Hooke's law) and $W = mg$ (weight)
- Resultant = $k \Delta x_{\text{max}} - mg$
- Resultant = $(560 \times 5) - (80 \times 9.81)$ [1 mark]
- Resultant = $2015.2 = 2020 \text{ N}$ [1 mark]

(ii) Calculate the extension of the rope when the climber's acceleration is zero:

- When acceleration = 0: $F = W$
- So, $mg = k \Delta x_{\text{min}} \Rightarrow \Delta x_{\text{min}} = \frac{mg}{k}$
- $\Delta x_{\text{min}} = \frac{80 \times 9.81}{560}$ [1 mark]
- $\Delta x_{\text{min}} = 1.4 \text{ m}$ [1 mark]

[Total: 5 marks]

It's useful to consider the free-body force diagrams of the climber at the different stages of his fall

At the highest point (before falling):

- The forces acting on the climber are his weight and the reaction force which are balanced
- Hence $R - mg = 0$

Between the highest point and L :

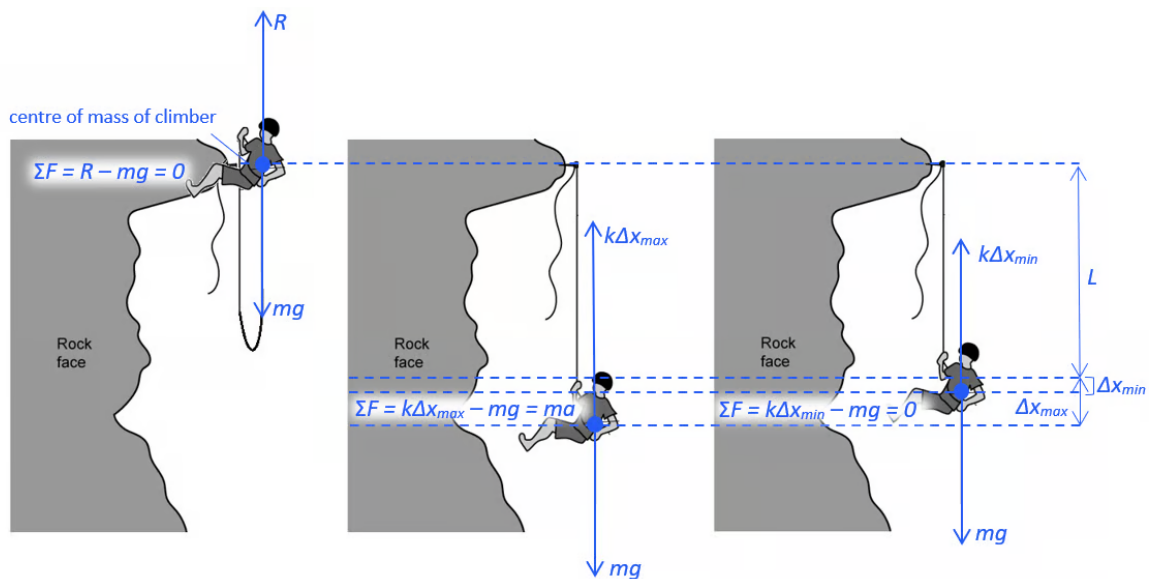
- The climber freefalls under gravity, only his weight acts (as air resistance/drag is ignored)

Between L and $\Delta x_{\max}$:

- The rope begins to stretch, and we are assuming extension is proportional to tension, so the restoring force is equal to Hooke's law
- Hence the resultant force is $k\Delta x_{\max} - mg = ma$
- $\Delta x_{\max}$ is the point where the climber comes to rest

Between $\Delta x_{\max}$ and $\Delta x_{\min}$:

- The extension decreases as the resultant force/acceleration decrease to zero
- $\Delta x_{\min}$ is the point where the climber comes to rest again and the forces acting on him are balanced again
- Hence $k\Delta x_{\min} - mg = 0$

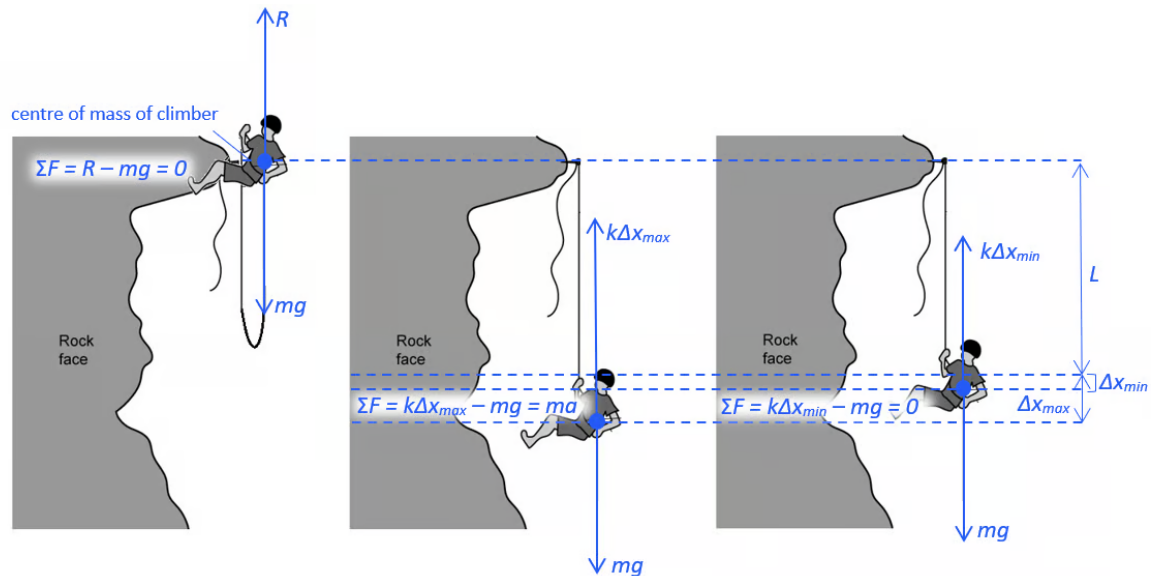


3b

(b)

(i) Calculate the maximum speed of the climber before the rope begins to stretch:

- Gain in KE = Change in GPE



3b

(b)

(i) Calculate the maximum speed of the climber before the rope begins to stretch:

- Gain in KE = Change in GPE
- $\frac{1}{2}mv^2 = mg\Delta h \Rightarrow v^2 = 2g\Delta h$
- $v = \sqrt{2g\Delta h} = \sqrt{2 \times 9.81 \times 19}$ [1 mark]
- Max speed: $v = 19.3 \text{ m s}^{-1}$ [1 mark]

(ii) If the climber used a longer rope:

- They would fall for longer and reach a higher velocity (just before the rope stretches); [1 mark]
- (Hence) the climber's deceleration / force (as the rope stretches) would be greater (not lower); [1 mark]

[Total: 4 marks]

3c

(c) Determine the average resultant forces exerted by the hemp and nylon ropes:

List the known quantities:

- Mass, $m = 80 \text{ kg}$
- Initial speed, $u = 19.3 \text{ m s}^{-1}$ (from **(b)**)
- Final speed, $v = 0$
- Time for nylon, $\Delta t_{\text{nylon}} = 0.664 \text{ s}$
- Time for hemp, $\Delta t_{\text{hemp}} = 0.035 \text{ s}$

Use impulse to determine the average resultant force exerted:

- Impulse: $F \Delta t = \Delta p \Rightarrow F = \frac{m \Delta v}{\Delta t}$
- Nylon: $\Delta F_{\text{nylon}} = \frac{80 \times 19.3}{0.664} = 2325 \text{ N}$ [1 mark]
- Hemp: $\Delta F_{\text{hemp}} = \frac{80 \times 19.3}{0.035} = 44\,114 \text{ N}$ [1 mark]

Determine the ratio of the forces:

- $\frac{\Delta F_{\text{hemp}}}{\Delta F_{\text{nylon}}} = \frac{44\,114}{2325} = 18.97$ [1 mark]
- Hence, the average resultant force exerted by hemp rope is about 20 times larger than that of the nylon rope

[Total: 3 marks]

3d

(d) Nylon ropes are favoured by rock climbers because:

Max **three** from:

- When a rock climber falls, they have a large momentum; [1 mark]

- The rope brings the climber to a stop / their momentum to zero; [1 mark]
- As the nylon rope stretches, the change in momentum occurs over a longer period of time; [1 mark]
- (This means that) for the same change in momentum, there is a smaller force applied to the climber (since force is the rate of change in momentum); [1 mark]
- (Hence) there is a lower risk of injury to the climber; [1 mark]

Compared to a hemp rope:

- A hemp rope would exert the same change in momentum over a shorter period of

3d

(d) Nylon ropes are favoured by rock climbers because:

Max **three** from:

- When a rock climber falls, they have a large momentum; [1 mark]
- The rope brings the climber to a stop / their momentum to zero; [1 mark]
- As the nylon rope stretches, the change in momentum occurs over a longer period of time; [1 mark]
- (This means that) for the same change in momentum, there is a smaller force applied to the climber (since force is the rate of change in momentum); [1 mark]
- (Hence) there is a lower risk of injury to the climber; [1 mark]

Compared to a hemp rope:

- A hemp rope would exert the same change in momentum over a shorter period of time;
AND
- (Therefore) it would apply a large force / injury to the climber; [1 mark]

[Total: 4 marks]